

Module Title: LAN Security Concepts

Module Objective: Explain how vulnerabilities compromise LAN security

Topic Title	Topic Objective
Endpoint Security	Explain how to use endpoint security to mitigate attacks
Access Control	Explain how AAA and 802.1x are used to authenticate LAN endpoints and devices
Layer 2 Security Threats	Identify Layer 2 vulnerabilities
MAC Address Table Attack	Explain how a MAC address table attack compromised LAN security
LAN Attacks	Explain how LAN attacks compromise LAN security

Page #	Activity Type	Activity Name
10.1.6	Check Your Understanding	Endpoint Security
10.2.7	Check Your Understanding	Access Control

10.3.4	Check Your Understanding	Layer 2 Security Threats
10.4.4	Check Your Understanding	MAC Address Table Attacks
10.5.1	Video	VLAN and DHCP Attacks
10.5.6	Video	ARP Attacks, STP Attacks, and CDP Reconnaissance
10.5.11	Check Your Understanding	LAN Attacks

New Terms

Data Breach

Malware

Next-Generation Firewall (NGFW)

Next-Generation IPS (NGIPS)

Advanced Malware Protection (AMP)

Authentication, Authorization, Accounting (AAA)

Identity Services Engine (ISE)

Host-Based Intrusion Prevention System (HIPS)

Email Security Appliance (ESA)

Web Security Appliance (WSA)

login {local}

Remote Authentication Dial-In User Service (RADIUS)

Terminal Access Controller Access Control System (TACACS+)

IEEE 802.1X

Client (Supplicant)

Authenticator

Port Security

DHCP Snooping

Dynamic ARP Inspection (DAI)

IP Source Guard (IPSG)

VLAN Hopping

VLAN Double-Tagging

DHCP Starvation

DHCP Spoofing

Gratuitous ARP

ARP Spoofing

ARP Poisoning

Cisco Discovery Protocol (CDP)

no cdp run

no cdp enable

no lldp run

no lldp transmit | receive